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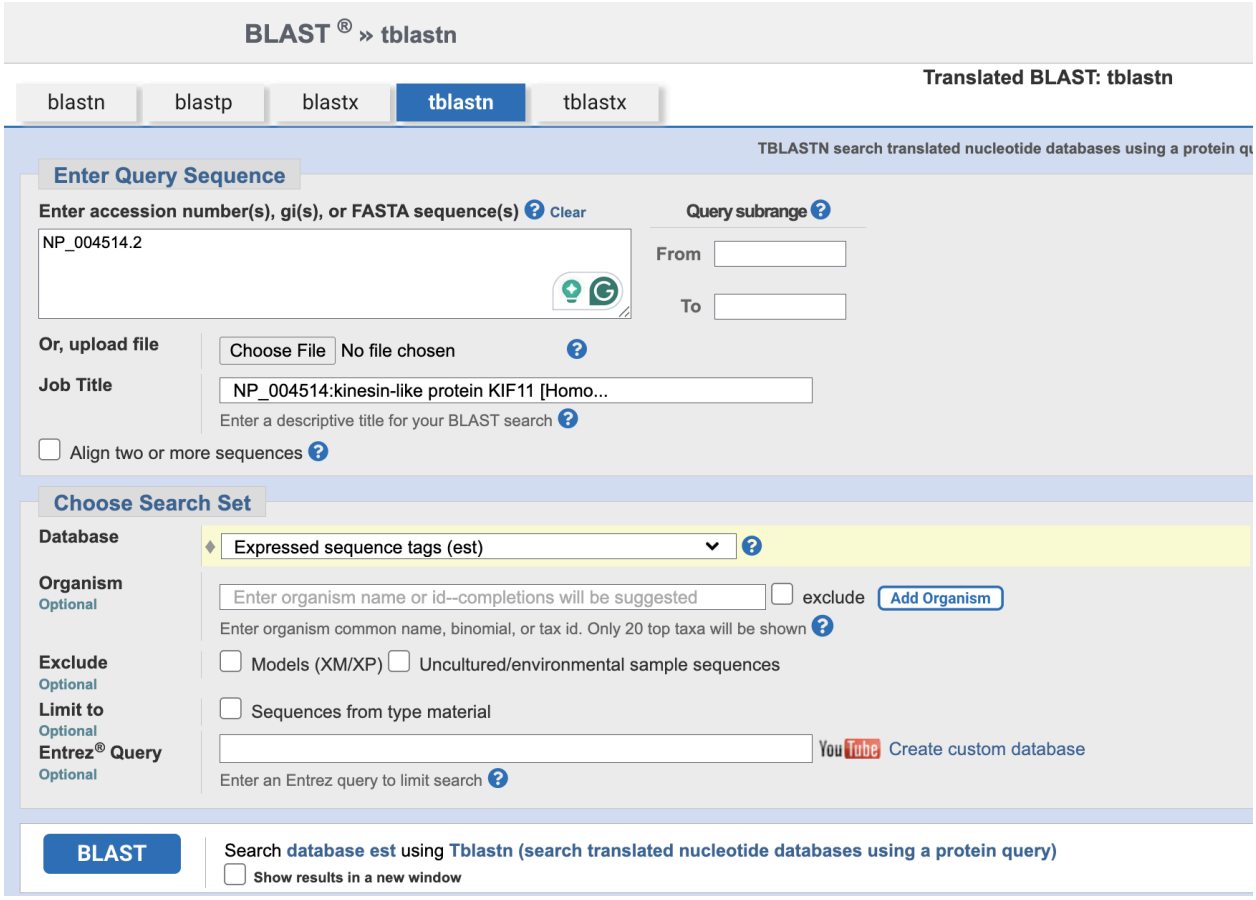
[Q1] Tell me the name of a protein you are interested in. Include the species, accession number and known function. This can be a human protein or a protein from any other species as long as it's function is known.
If you do not have a favorite protein, select human RBP4 or KIF11. Do not use beta globin as this is in the worked example report that I provide you with online.

Name: KIF11
Species: Homo sapiens
Accession: NP_004514.2
KIF11 is a motor protein that helps move microtubules during mitosis.

[Q2] Perform a BLAST search against a DNA database, such as a database consisting of genomic DNA or ESTs. The BLAST server can be at NCBI or elsewhere. Include details of the BLAST method used, database searched and any limits applied (e.g. Organism).

Method: TBLASTN search
Database: Expressed Sequence Tags (est)



Also include the output of that BLAST search in your document. If appropriate, change the font to Courier size 10 so that the results are displayed neatly. You can also screen capture a BLAST output (e.g. alt print screen on a PC or on a MAC press ⌘-shift-4. The pointer becomes a bulls eye. Select the area you wish to capture and release. The image is saved as a file called Screen Shot [].png in your Desktop directory). It is not necessary to print out all of the blast results if there are many pages.

Descriptions | Graphic Summary | Alignments | Taxonomy

Sequences producing significant alignments Download Select columns Show 100

select all *100 sequences selected* GenBank Graphics

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
HESC3_99_C12.g2_A036 NIH_MGC_260 Homo sapiens cDNA clone IMAGE:7484953 5', mRNA sequence	Homo sapiens	571	571	26%	0.0	100.00%	823	CX788535.1
1139617 MARC 4PIG Sus scrofa cDNA 3', mRNA sequence	Sus scrofa	556	556	27%	0.0	94.74%	858	DN125747.1
JGI_CAAV7594.fwd CAAV Pimephales promelas testis 7-8 month adults, males and females pooled (H) Pimeph...	Pimephales pro...	544	544	27%	0.0	87.80%	865	DT277589.1
AGENCOURT_6419004 NIH_MGC_71 Homo sapiens cDNA clone IMAGE:5535104 5', mRNA sequence	Homo sapiens	523	523	26%	4e-175	96.75%	1087	BM800579.1
HX442072 full-length enriched common marmoset ES cells cDNA library Callithrix jacchus cDNA clone MES-195...	Callithrix jacchus	518	518	26%	3e-174	95.99%	825	HX442072.1
HESC4_59_D04.g1_A037 NIH_MGC_262 Homo sapiens cDNA clone IMAGE:7486137 5', mRNA sequence	Homo sapiens	517	517	23%	5e-174	100.00%	828	CX871622.1
JGI_CAAN10013.fwd NIH_XGC_tropTe4 Xenopus tropicalis cDNA clone IMAGE:7695825 5', mRNA sequence	Xenopus tropicalis	516	516	25%	8e-174	90.87%	791	CX924637.2
HX934525 Muscle satellite cell of Korean cattle (Hanwoo), normalized cDNA library Bos taurus cDNA clone MSC...	Bos taurus	515	515	27%	4e-173	91.20%	867	HX934525.1
1100099 MARC 4PIG Sus scrofa cDNA 5', mRNA sequence	Sus scrofa	513	513	26%	2e-172	93.01%	819	DN104316.1
AGENCOURT_7973130 NIH_MGC_72 Homo sapiens cDNA clone IMAGE:6166881 5', mRNA sequence	Homo sapiens	512	512	25%	9e-172	93.73%	858	BU172266.1
AU120361 HEMBB1 Homo sapiens cDNA clone HEMBB1000527 5', mRNA sequence	Homo sapiens	509	509	25%	8e-171	96.63%	805	AU120361.1
JGI_XZG51134.fwd NIH_XGC_tropGas7 Xenopus tropicalis cDNA clone IMAGE:7566089 5', mRNA sequence	Xenopus tropicalis	496	496	25%	7e-166	86.74%	794	CX482778.2
HX934473 Muscle satellite cell of Korean cattle (Hanwoo), normalized cDNA library Bos taurus cDNA clone MSC...	Bos taurus	495	495	26%	1e-165	91.48%	817	HX934473.1
AGENCOURT_8878638 NIH_MGC_71 Homo sapiens cDNA clone IMAGE:6464974 5', mRNA sequence	Homo sapiens	495	495	24%	3e-164	93.16%	1086	BQ949030.1
AGENCOURT_10808106 MAPcL Homo sapiens cDNA clone IMAGE:6721154 5', mRNA sequence	Homo sapiens	493	493	22%	5e-164	100.00%	919	CA488948.1
UI-M-HO0-cpw-l-21-0-UI.r1 NIH_BMAP_HO0 Mus musculus cDNA clone IMAGE:30655676 5', mRNA sequence	Mus musculus	489	489	23%	8e-164	97.07%	722	CN533430.1
HX934316 Muscle satellite cell of Korean cattle (Hanwoo), normalized cDNA library Bos taurus cDNA clone MSC...	Bos taurus	483	483	25%	9e-161	91.25%	806	HX934316.1
UI-M-HS0-cqe-j-10-0-UI.r1 NIH_BMAP_HS0 Mus musculus cDNA clone IMAGE:30668673 5', mRNA sequence	Mus musculus	481	481	24%	1e-160	94.80%	752	CN534676.1
JGI_ANNO41985.fwd ANNO Pimephales promelas Whole (M) Pimephales promelas cDNA clone ANNO41985 5'...	Pimephales pro...	468	468	25%	2e-155	85.77%	783	DT170354.1
FS309663 Cp_aO Cynops pyrrhogaster cDNA clone Cp_aO_024_L08 3', mRNA sequence	Cynops pyrrhoga...	424	577	32%	3e-155	84.36%	1055	FS309663.1
UI-M-FY0-cdr-o-02-0-UI.r1 NIH_BMAP_FY0 Mus musculus cDNA clone IMAGE:6833595 5', mRNA sequence	Mus musculus	468	468	23%	3e-155	93.44%	743	CB244717.1

On the BLAST results, clearly indicate a match that represents a protein sequence, encoded from some DNA sequence, that is homologous to your query protein. I need to be able to inspect the pairwise alignment you have selected, including the E value and score. It should be labeled a "genomic clone" or "mRNA sequence", etc. - but include no functional annotation.

Chosen match: Accession FS309663.1, a 1055 base pair clone from *Cynops pyrrhogaster.*

Edit Search | Save Search | Search Summary | How to read this report? | BLAST Help Videos | Back to Traditional Results Page

Job Title	NP_004514:kinesin-like protein KIF11 [Homo...
RID	S32DZ4CV014 *Search expires on 02-04 12:38 pm* Download All
Program	TBLASTN Citation
Database	est See details
Query ID	NP_004514.2
Description	kinesin-like protein KIF11 [Homo sapiens]
Molecule type	amino acid
Query Length	1056
Other reports	

Filter Results

Organism *only top 20 will appear* exclude

Type common name, binomial, taxid or group name

+ Add organism

Percent Identity to | E value to | Query Coverage to

Filter | Reset

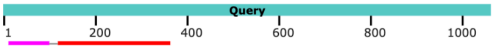
Descriptions | **Graphic Summary** | Alignments | Taxonomy

hover to see the title *click to show alignments*

Alignment Scores < 40 | 40 - 50 | 50 - 80 | 80 - 200 | >= 200

1 sequences selected

Distribution of the top 2 Blast Hits on 1 subject sequences



Alignment details:

Descriptions | Graphic Summary | **Alignments** | Taxonomy

Alignment view Pairwise | Restore defaults | Download

1 sequences selected

Download | GenBank | Graphics | Sort by: E value | Next | Previous | Descriptions

FS309663 Cp_aO Cynops pyrrhogaster cDNA clone Cp_aO_024_L08 3', mRNA sequence

Sequence ID: FS309663.1 Length: 1055 Number of Matches: 2

Range 1: 326 to 1054 GenBank Graphics | Next Match | Previous Match

Related Information

Score	Expect	Method	Identities	Positives	Gaps	Frame
424 bits(1091)	3e-155	Compositional matrix adjust.	212/243(87%)	230/243(94%)	0/243(0%)	+2

```
Query  120   SPNEEYTWEEDPLAGIIPRTLHQIFEKLTDNGTEFSVKVSLLEIYNEELFDLLNPSSDVS  179
             +P+ E+TW+EDPLAGIIPRTLHQIFEKL++NGTEFSVKVSLLEIYNEELFDLL+P  DV+
Sbjct  326   APDGEFTWDEDPLAGIIPRTLHQIFEKLSENGTEFSVKVSLLEIYNEELFDLLSPCPDVN  505

Query  180   ERLQMFDDPRNKRGVIIKGLEEITVHNKDEVYQILEKGAAKRTTAATLMNAYSSRSHSVF  239
             ERLQMFDDPRNKRGVIIKGLEE+TVHNK+EVYQILE+GAAKRTTA+TLMNAYSSRSHSVF
Sbjct  506   ERLQMFDDPRNKRGVIIKGLEEVTVHNKNEVYQILERGAAKRTTASTLMNAYSSRSHSVF  685

Query  240   SVTIHMKETTIDGEELVKIGKLNLVDLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVI  299
             SVTIHMKETT+DGEELVK GKLNLV LAGSENIGRSGAVDKRAREAGNINQSLLTLGRVI
Sbjct  686   SVTIHMKETTVDGEELVKXGKLNLVXLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVI  865

Query  300   TALVERTPHVPYRESKLTRILQDSLGGRTRTSIIATISPAslnleetlstleYAHRAKNI  359
             TAL ERTPH+PYR SKLTRILQ SLG  T+TSIIAT SPAS+N EETLSTL YA+RAKNI
Sbjct  866   TALXERTPHIPYRXSKLTRILQXSLGXXTKTSIIATXSPASINXEETLSTLXYAYRAKNI  1045

Query  360   LNK  362
             +NK
Sbjct  1046  MNK  1054
```

Range 2: 61 to 330 GenBank Graphics | Next Match | Previous Match | First Match

Score	Expect	Method	Identities	Positives	Gaps	Frame
152 bits(385)	3e-155	Composition-based stats.	66/90(73%)	80/90(88%)	0/90(0%)	+1

```
Query  11   KKEEKGKNIQVVVRCRPFNLAERKASAHSIVECDPVRKEVSVRTGGLADKSSRKTYTFDM  70
            KK++KGKNIQVVVRCRPFNL ERK  ++S+++C+P RKE++VR GG+ +K  RKTYTFDM
Sbjct  61   KKDDKGKNIQVVVRCRPFNLQERKGGSYSVIDCEPSRKEIAVRPGGINEKIGRKTYTFDM  240

Query  71   VFGASTKQIDVYRSVVCPILDEVIMGYNCT  100
            VFG S KQI+VYR+VVCPILDEVIMGYNCT
Sbjct  241  VFGPSAKQIEVYRNVVCPILDEVIMGYNCT  330
```

In general, [Q2] is the most difficult for students because it requires you to have a “feel” for how to interpret BLAST results. You need to distinguish between a perfect match to your query (i.e. a sequence that is not “novel”), a near match (something that might be “novel”, depending on the results of [Q4]), and a non-homologous result.
If you are having trouble finding a novel gene try restricting your search to an organism that is poorly annotated.

[Q3] Gather information about this “novel” protein. At a minimum, show me the protein sequence of the “novel” protein as displayed in your BLAST results from [Q2] as FASTA format (you can copy and paste the aligned sequence subject lines from your BLAST result page if necessary) or translate your novel DNA sequence using a tool called EMBOSS Transeq at the EBI. Don’t forget to translate all six reading frames; the ORF (open reading frame) is likely to be the longest sequence without a stop codon. It may not start with a methionine if you don’t have the complete coding region. Make sure the sequence you provide includes a header/subject line and is in traditional FASTA format.

>EMBOSS_001_1: pyrrhogaster protein (Japanese fire-bellied newt)
APDGEFTWDEDPLAGIIPRTLHQIFEKLSENGTEFSVKVSLLEIYNEELFDLLSPCPDVN
ERLQMFDDPRNKRGVIIKGLEEVTVHNKNEVYQILERGAAKRTTASTLMNAYSSRSHSVF
SVTIHMKETTVDGEELVKXGKLNLVXLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVI
TALXERTPHIPYRXSKLTRILQXSLGXXTKTSIIATXSPASINXEETLSTLXYAYRAKNI
MNK

Here, tell me the name of the novel protein, and the species from which it derives. It is very unlikely (but still definitely possible) that you will find a novel gene from an organism such as S. cerevisiae, human or mouse, because those genomes have already been thoroughly annotated. It is more likely that you will discover a new gene in a genome that is currently being sequenced, such as bacteria or plants or protozoa.

Name: FS309663 Cp_aO Cynops pyrrhogaster cDNA clone Cp_aO_024_L08 3', mRNA sequence.
Species: *Cynops* pyrrhogaster
Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Amphibia; Batrachia; Caudata; Salamandroidea; Salamandridae; Pleurodelinae; Cynops.

[Q4] Prove that this gene, and its corresponding protein, are novel. For the purposes of this project, “novel” is defined as follows. Take the protein sequence (your answer to [Q3]), and use it as a query in a blastp search of the nr database at NCBI.

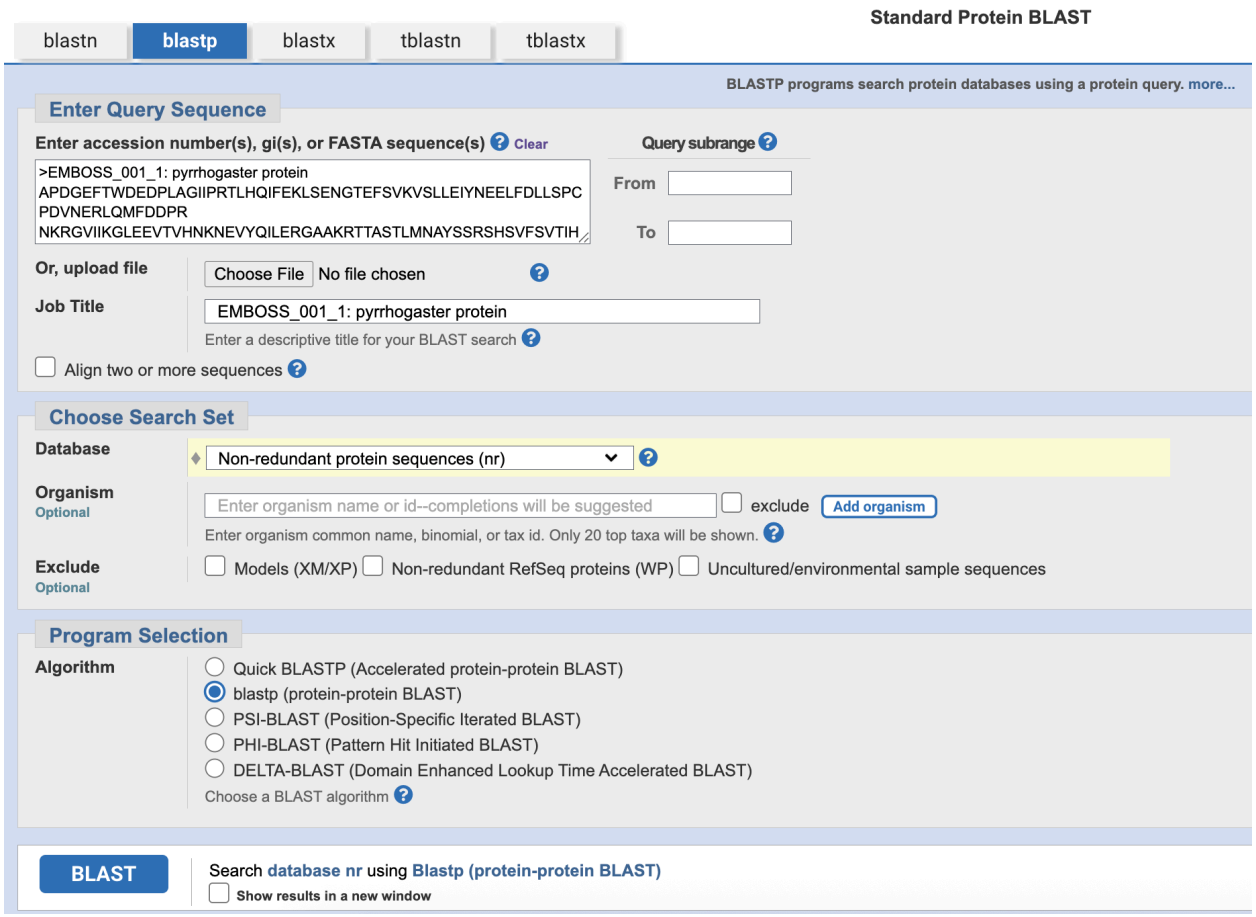
• If there is a match with 100% amino acid identity to a protein in the database, from the same species, then your protein is NOT novel (even if the match is to a protein with a name such as “unknown”). Someone has already found and annotated this sequence, and assigned it an accession number.

• If the top match reported has less than 100% identity, then it is likely that your protein is novel, and you have succeeded.

• If there is a match with 100% identity, but to a different species than the one you started with, then you have likely succeeded in finding a novel gene.

• If there are no database matches to the original query from [Q1], this indicates that you have partially succeeded: yes, you may have found a new gene, but no, it is not actually homologous to the original query. You should probably start over.

Details:
A BLASTP search against NR database yielded a top hit result to a protein from *Pleurodeles waltl* (Iberian ribbed newt).



The top result is a protein from *Pleurodeles waltl* (Iberian ribbed newt).

Descriptions | Graphic Summary | Alignments | Taxonomy

Sequences producing significant alignments Download Select columns Show 100

select all *100 sequences selected* GenPept Graphics Distance tree of results Multiple alignment MSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
hypothetical protein NDU88_007895 [Pleurodeles waltl]	Pleurodeles waltl	479	479	100%	7e-159	95.06%	1082	KAJ1141567.1
kinesin-like protein KIF11 [Pleurodeles waltl]	Pleurodeles waltl	479	479	100%	8e-159	95.06%	1070	XP_069095912.1
kinesin-like protein KIF11 [Rhinatrema bivittatum]	Rhinatrema bivittatum	454	454	100%	1e-157	91.36%	391	XP_029432074.1
kinesin-like protein KIF11 [Lissotriton helveticus]	Lissotriton helveticus	475	475	100%	2e-157	94.65%	1070	XP_078499533.1
kinesin-like protein KIF11 [Ambystoma mexicanum]	Ambystoma mexicanum	475	475	100%	3e-157	94.24%	1066	XP_069467469.1
hypothetical protein EK904_009114 [Melospiza melodia maxima]	Melospiza melodia maxima	449	449	100%	2e-155	89.71%	415	KAF2975241.1
kinesin-like protein KIF11 [Discoglossus pictus]	Discoglossus pictus	463	463	100%	7e-153	91.77%	1069	KAM4636768.1

Descriptions | Graphic Summary | Alignments | Taxonomy

Alignment view Pairwise Restore defaults Download

2 sequences selected

Download GenPept Graphics ▼Next ▲Previous ◀◀Descriptions

hypothetical protein NDU88_007895 [Pleurodeles waltl]

Sequence ID: KAJ1141567.1 **Length:** 1082 **Number of Matches:** 1

Range 1: 132 to 374 GenPept Graphics ▼Next Match ▲Previous Match

Score	Expect	Method	Identities	Positives	Gaps
479 bits(1234)	7e-159	Compositional matrix adjust.	231/243(95%)	233/243(95%)	0/243(0%)

```
Query  1    APDGEFTWDEDPLAGIIPRTLHQIFEKLSENGTEFSVKVSLLEIYNEELFDLLSPCPDVN  60
            +PDGEFTWDEDPLAGIIPRTLHQIFEKLSENGTEFSVKVSLLEIYNEELFDLLSPCPDVN
Sbjct  132  SPDGEFTWDEDPLAGIIPRTLHQIFEKLSENGTEFSVKVSLLEIYNEELFDLLSPCPDVN  191

Query  61   ERLQMFDDPRNKRGVIIKGLEEVTVHNKNEVYQILERGAAKRTTASTLMNAYSSRSHSVF  120
            ERLQMFDDPRNKRGVIIKGLEEVTVHNKNEVYQILERGAAKRTTASTLMNAYSSRSHSVF
Sbjct  192  ERLQMFDDPRNKRGVIIKGLEEVTVHNKNEVYQILERGAAKRTTASTLMNAYSSRSHSVF  251

Query  121  SVTIHMKETTVDGEELVKXGKLNLVXLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVI  180
            SVTIHMKETTVDGEELVK GKLNLV LAGSENIGRSGAVDKRAREAGNINQSLLTLGRVI
Sbjct  252  SVTIHMKETTVDGEELVKIGKLNLVDLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVI  311

Query  181  TALXERTPHIPYRXSKLTRILQXSLGXXTKTSIIATXSPASINXEETLSTLXYAYRAKNI  240
            TAL ERTPHIPYR SKLTRILQ SLG  TKTSIIAT SPASIN EETLSTL YA+RAKNI
Sbjct  312  TALVERTPHIPYRESKLTRILQDSLGGRTKTSIIATVSPASINLEETLSTLDYAHRAKNI  371

Query  241  MNK  243
            MNK
Sbjct  372  MNK  374
```

Related Information

Genome Data Viewer - aligned genomic context

[Q5] Generate a multiple sequence alignment with your novel protein, your original query protein, and a group of other members of this family from different species. A typical number of proteins to use in a multiple sequence alignment for this assignment purpose is a minimum of 5 and a maximum of 20 - although the exact number is up to you. Include the multiple sequence alignment in your report. Use Courier font with a size appropriate to fit page width.
Side-note: Indicate your sequence in the alignment by choosing an appropriate name for each sequence in the input unaligned sequence file (i.e. edit the sequence file so that the species, or short common, names (rather than accession numbers) display in the output alignment and in the subsequent answers below). The goal in this step is to create an interesting an alignment for building a phylogenetic tree that illustrates species divergence.

Re-labeled sequences for alignment

>EMBOSS_001_1: pyrrhogaster protein (sequence taken from BLAST result)
APDGEFTWDEDPLAGIIPRTLHQIFEKLSENGTEFSVKVSLLEIYNEELFDLLSPCPDVN
ERLQMFDDPRNKRGVIIKGLEEVTVHNKNEVYQILERGAAKRTTASTLMNAYSSRSHSVF
SVTIHMKETTVDGEELVKXGKLNLVXLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVI
TALXERTPHIPYRXSKLTRILQXSLGXXTKTSIIATXSPASINXEETLSTLXYAYRAKNI
MNK

>NP_004514.2 kinesin-like protein KIF11 [Homo sapiens]
SPNEEYTWEEDPLAGIIPRTLHQIFEKLTDNGTEFSVKVSLLEIYNEELFDLLNPSSDVSERLQMF
DDPRNKRGVIIKGLEEITVHNKDEVYQILEKGAAKRTTAATLMNAYSSRSHSVFSVTIHMKETTID
GEELVKIGKLNLVDLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVITALVERTPHVPYRESKLT
RILQDSLGGRTRTSIIATISPASLNLEETLSTLEYAHRAKNILNK

>XP_069095912.1 kinesin-like protein KIF11 [Pleurodeles waltl]
SPDGEFTWDEDPLAGIIPRTLHQIFEKLSENGTEFSVKVSLLEIYNEELFDLLSPCPDVNERLQM
FDDPRNKRGVIIKGLEEVTVHNKNEVYQILERGAAKRTTASTLMNAYSSRSHSVFSVTIHMKETT
VDGEELVKIGKLNLVDLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVITALVERTPHIPYRESK
LTRILQDSLGGRTKTSIIATVSPASINLEETLSTLDYAHRAKNIMNK

>XP_069467469.1 kinesin-like protein KIF11 [Ambystoma mexicanum]
SPDGEFTWDEDPLAGIIPRTLHQIFEKLSENGTEFSVKVSLLEIYNEELFDLLSPCPDVGERLQM
FDDPRNKRGVIIKGLEEITVHNKNEVYQILERGAAKRTTASTLMNAYSSRSHSVFSVTIHMKETT
VDGEELVKIGKLNLVDLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVITALVERTPHIPYRESK
LTRILQDSLGGRTKTSIIATVSPASINLEETLSTLEYAHRAKNIMNK

>XP_065264224.1 kinesin-like protein KIF11 [Emys orbicularis]
SPNEEYTWEEDPLAGIIPRTLHQIFEKLSDNGTEFSVKVSLLEIYNEELFDLLNPSPDVGERLQM
FDDPRNKRGVIIKGLEEITVHNKDEVYQILERGAAKRTTAATYMNAYSSRSHSVFSVTIHMKETTI

DGEELVKIGKLNLVDLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVITALVERTPHIPYRESKL
TRILQDSLGGRTKTSIIATVSPASVNLEETLSTLEYAHRAKNIMNK

>XP_029554436.1 kinesin-like protein KIF11 [Salmo trutta]
SPNEEFTWEEDPLAGIIPRTLHQIFEKLSENGTEFSVKVSLLEIYNEELFDLLSPCPDVTERLQLF
DDPRNKRGVIIKGLEEITVHNKNEVYQILERGSAKRKTASTLMNAYSSRSHSVFSVTIHMKEITIE
GEELVKIGKLNLVDLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVITALVEKRPHVPYRESKLT
RILQDSLGGRTKTSIIATVSPASINLEETLSTLEYANRAKNIMNK

>XP_054133353.1 kinesin-like protein KIF11 [Melozone crissalis]
SPNEEYTWEEDPLAGIIPRTLHQIFEKLTENGTEFSVKVSLLEIYNEELFDLLNPTPDVGERLQMF
DDPRNKRGVIIKGLEEVTVHNKNQVYQILERGAAKRTTAATYMNAYSSRSHSVFSITIHMKETTV
DGEELVKIGKLNLVDLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVITALVERAPHIPYRESKL
TRILQDSLGGRTKTSIIATISPASVNLEETLSTLEYAHRAKNIMNK

>XP_041591311.1 kinesin-like protein KIF11 [Vulpes lagopus]
SPNEEYTWEEDPLAGIIPRTLHQIFEKLTDNGTEFSVKVSLLEIYNEELFDLLNPSSDVSERLQMF
DDPRNKRGVIIKGLEEVTVHNKDEVYQILEKGAAKRTTAATLMNAYSSRSHSVFSVTIHMKETTI
DGEELVKIGKLNLVDLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVITALVERTPHVPYRESKL
TRILQDSLGGRTRTSIIATISPASLNLEETLSTLEYAHRAKNILNK

>XP_040110463.1 kinesin-like protein KIF11 [Oryx dammah]
SPNEEYTWEEDPLAGIIPRTLHQIFEKLTDNGTEFSVKVSLLEIYNEELFDLLNPSSDVSERLQMF
DDPRNKRGVIIKGLEEITVHNKDEVYQILEKGAAKRTTAATLMNAYSSRSHSVFSVTIHMKETTV
DGEELVKIGKLNLVDLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVITALVERTPHVPYRESKL
TRILQDSLGGRTRTSIIATVSPASLNLEETLSTLEYAHRAKNILNK

Alignment:
Obtained using MUSCLE (version 3.8) at EBI:
(The formatting was off when I copy and pasted so I attached screenshots instead)

```
CLUSTAL multiple sequence alignment by MUSCLE (3.8)


Brown_trout                     SPNEEFTWEEDPLAGIIPRTLHQIFEKLSENGTEFSVKVSLLEIYNEELFDLLSPCPDVT
California_towhee               SPNEEYTWEEDPLAGIIPRTLHQIFEKLTENGTEFSVKVSLLEIYNEELFDLLNPTPDVG
European_pond_turtle            SPNEEYTWEEDPLAGIIPRTLHQIFEKLSDNGTEFSVKVSLLEIYNEELFDLLNPSPDVG
Scimitar_oryx                   SPNEEYTWEEDPLAGIIPRTLHQIFEKLTDNGTEFSVKVSLLEIYNEELFDLLNPSSDVS
Human                           SPNEEYTWEEDPLAGIIPRTLHQIFEKLTDNGTEFSVKVSLLEIYNEELFDLLNPSSDVS
Arctic_fox                      SPNEEYTWEEDPLAGIIPRTLHQIFEKLTDNGTEFSVKVSLLEIYNEELFDLLNPSSDVS
Japanese_fire-bellied_newt      APDGEFTWDEDPLAGIIPRTLHQIFEKLSENGTEFSVKVSLLEIYNEELFDLLSPCPDVN
Iberian_ribbed_newt             SPDGEFTWDEDPLAGIIPRTLHQIFEKLSENGTEFSVKVSLLEIYNEELFDLLSPCPDVN
Axolotl                         SPDGEFTWDEDPLAGIIPRTLHQIFEKLSENGTEFSVKVSLLEIYNEELFDLLSPCPDVG
                                :*: *:**:********************::************************.* .**

Brown_trout                     ERLQLFDDPRNKRGVIIKGLEEITVHNKNEVYQILERGSAKRKTASTLMNAYSSRSHSVF
California_towhee               ERLQMFDDPRNKRGVIIKGLEEVTVHNKNQVYQILERGAAKRTTAATYMNAYSSRSHSVF
European_pond_turtle            ERLQMFDDPRNKRGVIIKGLEEITVHNKDEVYQILERGAAKRTTAATYMNAYSSRSHSVF
Scimitar_oryx                   ERLQMFDDPRNKRGVIIKGLEEITVHNKDEVYQILEKGAAKRTTAATLMNAYSSRSHSVF
Human                           ERLQMFDDPRNKRGVIIKGLEEITVHNKDEVYQILEKGAAKRTTAATLMNAYSSRSHSVF
Arctic_fox                      ERLQMFDDPRNKRGVIIKGLEEVTVHNKDEVYQILEKGAAKRTTAATLMNAYSSRSHSVF
Japanese_fire-bellied_newt      ERLQMFDDPRNKRGVIIKGLEEVTVHNKNEVYQILERGAAKRTTASTLMNAYSSRSHSVF
Iberian_ribbed_newt             ERLQMFDDPRNKRGVIIKGLEEVTVHNKNEVYQILERGAAKRTTASTLMNAYSSRSHSVF
Axolotl                         ERLQMFDDPRNKRGVIIKGLEEITVHNKNEVYQILERGAAKRTTASTLMNAYSSRSHSVF
                                ****:*****************:*****::******.*:***.**:* ************
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Brown_trout                     SVTIHMKEITIEGEELVKIGKLNLVDLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVI
California_towhee               SITIHMKETTVDGEELVKIGKLNLVDLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVI
European_pond_turtle            SVTIHMKETTIDGEELVKIGKLNLVDLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVI
Scimitar_oryx                   SVTIHMKETTVDGEELVKIGKLNLVDLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVI
Human                           SVTIHMKETTIDGEELVKIGKLNLVDLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVI
Arctic_fox                      SVTIHMKETTIDGEELVKIGKLNLVDLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVI
Japanese_fire-bellied_newt      SVTIHMKETTVDGEELVKXGKLNLVXLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVI
Iberian_ribbed_newt             SVTIHMKETTVDGEELVKIGKLNLVDLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVI
Axolotl                         SVTIHMKETTVDGEELVKIGKLNLVDLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVI
                                *:****** *::****** ****** **********************************

Brown_trout                     TALVEKRPHVPYRESKLTRILQDSLGGRTKTSIIATVSPASINLEETLSTLEYANRAKNI
California_towhee               TALVERAPHIPYRESKLTRILQDSLGGRTKTSIIATISPASVNLEETLSTLEYAHRAKNI
European_pond_turtle            TALVERTPHIPYRESKLTRILQDSLGGRTKTSIIATVSPASVNLEETLSTLEYAHRAKNI
Scimitar_oryx                   TALVERTPHVPYRESKLTRILQDSLGGRTRTSIIATVSPASLNLEETLSTLEYAHRAKNI
Human                           TALVERTPHVPYRESKLTRILQDSLGGRTRTSIIATISPASLNLEETLSTLEYAHRAKNI
Arctic_fox                      TALVERTPHVPYRESKLTRILQDSLGGRTRTSIIATISPASLNLEETLSTLEYAHRAKNI
Japanese_fire-bellied_newt      TALXERTPHIPYRXSKLTRILQXSLGXXTKTSIIATXSPASINXEETLSTLXYAYRAKNI
Iberian_ribbed_newt             TALVERTPHIPYRESKLTRILQDSLGGRTKTSIIATVSPASINLEETLSTLDYAHRAKNI
Axolotl                         TALVERTPHIPYRESKLTRILQDSLGGRTKTSIIATVSPASINLEETLSTLEYAHRAKNI
                                *** *. **:*** ******** ***  *.****** ****:* ******* ** *****

Brown_trout                     MNK
California_towhee               MNK
European_pond_turtle            MNK
Scimitar_oryx                   LNK
Human                           LNK
Arctic_fox                      LNK
Japanese_fire-bellied_newt      MNK
Iberian_ribbed_newt             MNK
Axolotl                         MNK
                                :**
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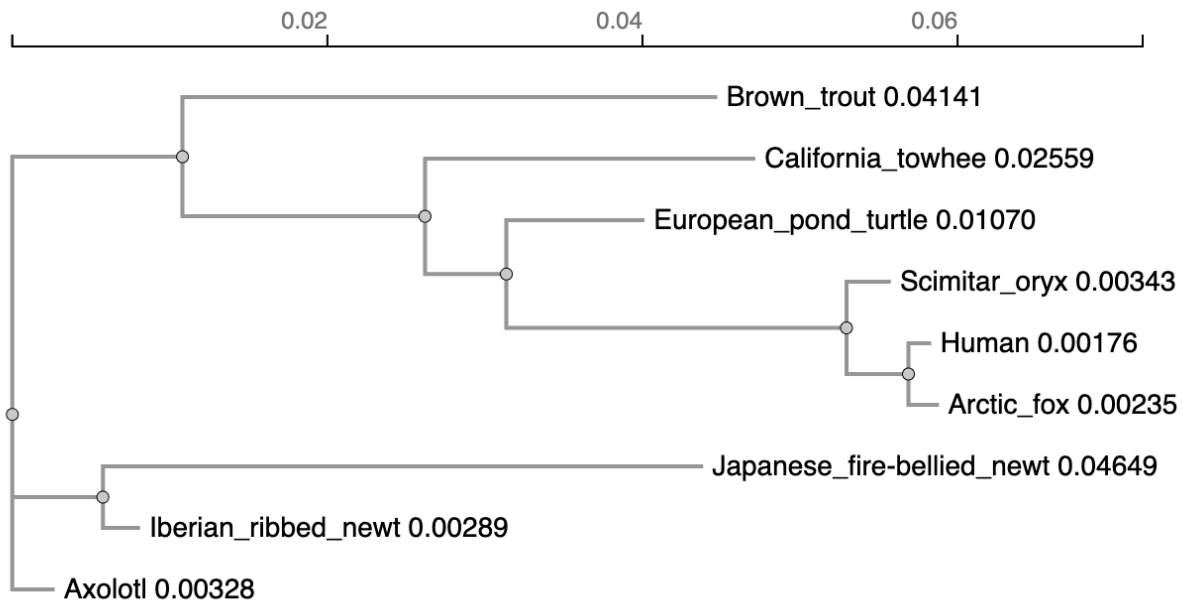
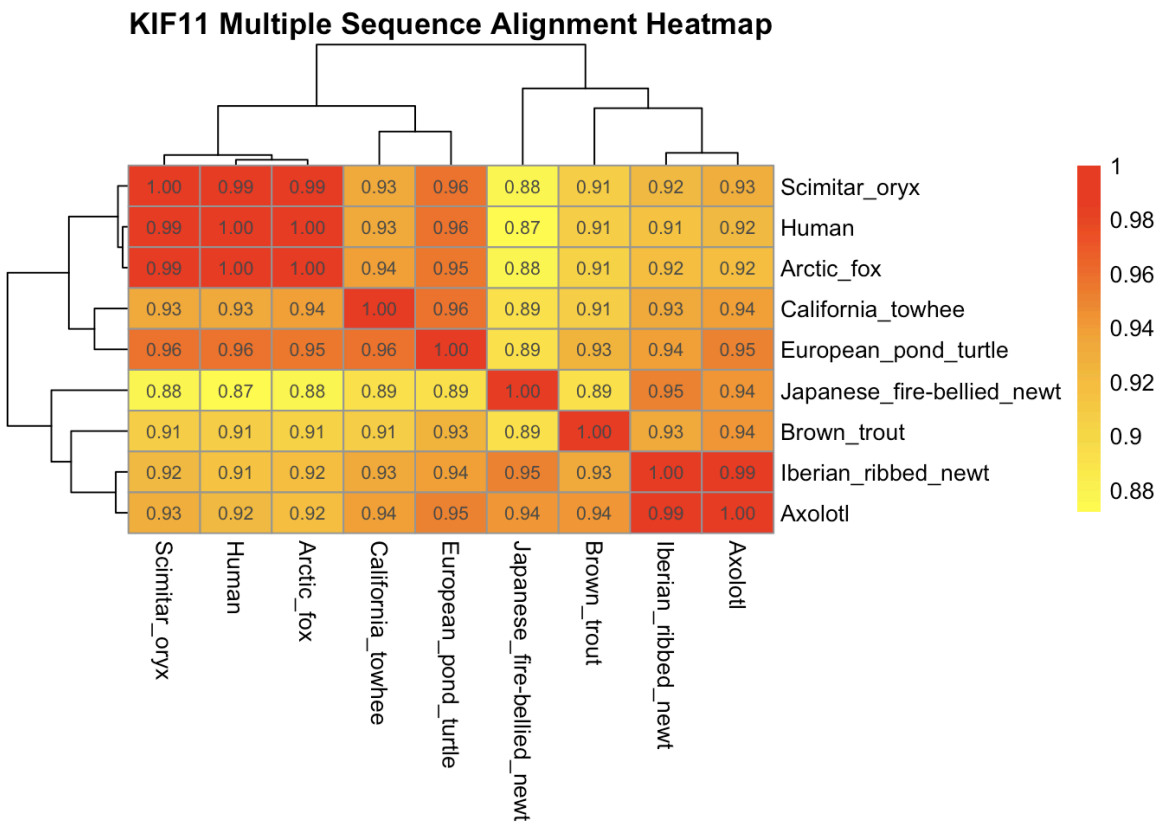
[Q6] Create a phylogenetic tree, using either a parsimony or distance-based approach. Bootstrapping and tree rooting are optional. Use “simple phylogeny” online from the EBI or any respected phylogeny program (such as MEGA, PAUP, or Phylip). Paste an image of your Cladogram or tree output in your report.



[Q7] Generate a sequence identity based heatmap of your aligned sequences using R. If necessary convert your sequence alignment to the ubiquitous FASTA format (Seaview can read in clustal format and “Save as” FASTA format for example). Read this FASTA format alignment into R with the help of functions in the Bio3D package. Calculate a sequence identity matrix (again using a function within the Bio3D package). Then generate a heatmap plot and add to your report. Do make sure your labels are visible and not cut at the figure margins.

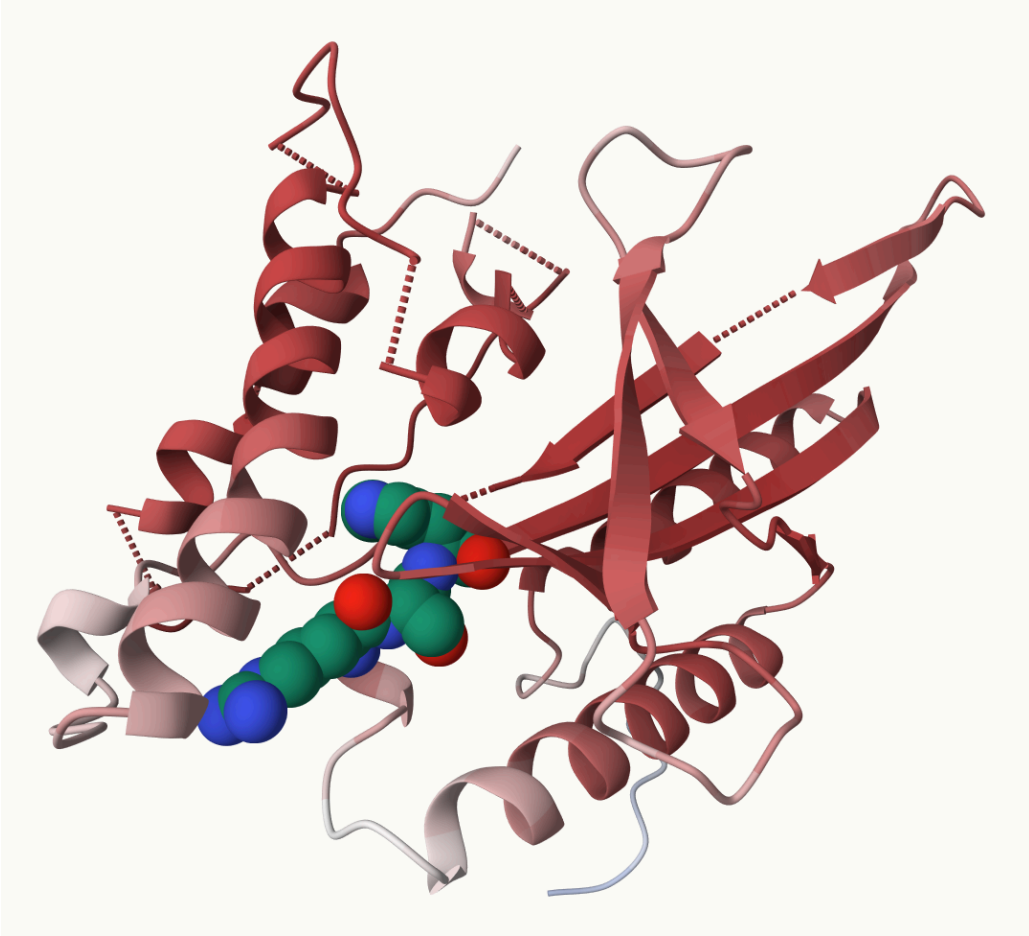


[Q8] Using R/Bio3D (or an online blast server if you prefer), search the main protein structure database for the most similar atomic resolution structures to your aligned sequences.
List the top 3 unique hits (i.e. not hits representing different chains from the same structure) along with their Evalue and sequence identity to your query. Please also add annotation details of these structures. For example include the annotation terms PDB identifier (structureId), Method used to solve the structure (experimentalTechnique), resolution (resolution), and source organism (source).

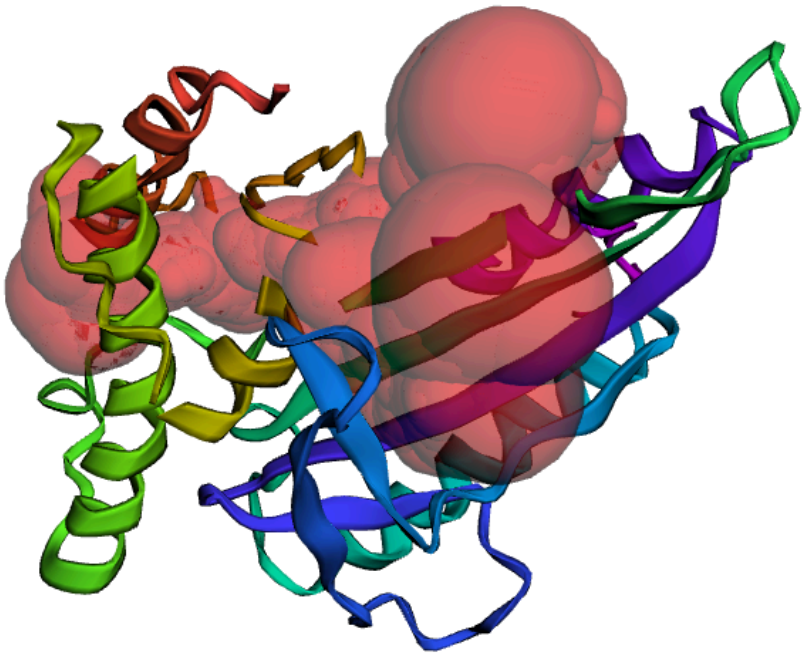
ID	Technique	Resolution	Source	E-value	Identity
8YHH	X-ray diffraction	1.950	Homo sapiens	1.16e-145	93.004
6VPO	Electron microscopy	4.400	Drosophila melanogaster	3.22e-87	61.864
5MM4	Electron microscopy	4.500	Mycosarcoma maydis	3.46e-76	56.327

[Q9] Using AlphaFold notebook generate a structural model using the default parameters for your novel protein sequence.
Note that this can take some time depending upon your sequence length. If your model is taking many hours to generate or your input sequence yields a “too many amino acids” (i.e. length) error you can focus on a single domain from your sequence - identify region by searching for PFAM domain matches.
Once complete save the resulting PDB format file for your records. Finally, generate a molecular figure of your generated PDB structure using the Mol* viewer online (or VMD/PyMol/Chimera if you prefer). To complete your analysis you should highlight conserved residues that are likely to be functional as spacefill and the protein as cartoon colored by local alpha fold pLDDT quality score. You can determine conserved residues from the alignment generated by the AlphaFold server and use a conservation cutoff appropriate for the diversity of your protein alignment (e.g. between 60% and 99% conserved). Note that pLDDT score is contained in the B-factor column of your PDB downloaded file. Please use a white or transparent background for your figure (i.e. not the default black in PyMol/VMD/Chimera etc.).

The conservation cutoff was 88%.



[Q10] (i) Using your computed structure model (or your closest homologue of known structure from the PDB) predict and locate potential small molecule binding sites using the CASTpFold server (https://cfold.bme.uic.edu/castpfold/). Provide an image or screen-shot of your largest predicted pockets “negative volume” and provide it’s area and volume.



Area = 1369.698 $Å^2$
Volume = 1685.028 $Å^3$

(ii) Perform a “Target” search of ChEMBEL (https://www.ebi.ac.uk/chembl/) with your novel sequence. Are there any Target Associated Assays and ligand efficiency data reported that may be useful starting points for exploring potential inhibition of your novel protein? If there are no assays listed here simply list “non available as of [date]”.

None available as of 03/10/2026

(iii) Briefly discuss (100 words max) the druggability of your novel protein based on:
- Presence of well-defined pockets (output of tools like CASTpFold),
- Existence of known inhibitors for related proteins (your search of ChEMBEL),

- Conservation of binding sites across homologs (your conservation analysis in Q10),
- Potential therapeutic applications if this protein were targeted (you can use ChatGPT, Claude etc. backed up by your reading of the literature here).

The largest pocket and the most conserved region of the protein are in two separate locations. The most conserved binding site does not have a pocket, so drugs cannot bind to that region. Therefore, it would be difficult to have a drug target this protein because there is no common region across all homologs where a drug could bind. Furthermore, there were no results from ChEMBEL, which means that currently there are no known drugs that inhibit this protein.